

Validation packet control and method scope

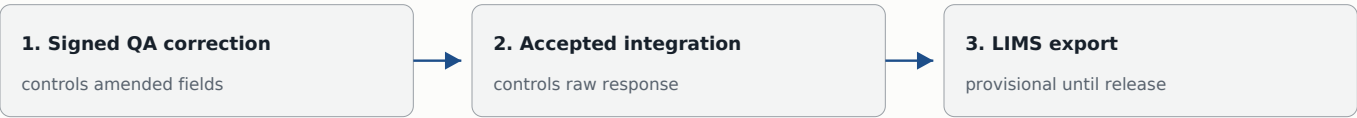
North River drinking-water study | EPA 533 modified | Revision B

CONTROLLED COPY

CONTROLLED METHOD RECORD

Field	Controlled value
Study	MV-PFAS-24-017
Instrument	SCIEX 6500+ QTRAP; ESI negative
Column	BEH C18, 2.1 x 50 mm, 1.7 um
Injection	5 uL
Study interval	08-22 Apr 2024
Reporting basis	ng/L in original sample; U95 where quantified

SOURCE PRECEDENCE



BOTTLE MANIFEST

Sample ID	Matrix	Collected	Received	Preserved	Release path
FB-240408-01	Field blank	08 Apr 09:10	08 Apr 15:42	Trizma	Routine
DW-240408-02	Finished water	08 Apr 09:25	08 Apr 15:42	Trizma	Routine
DW-240408-03	Finished water	08 Apr 09:40	08 Apr 15:42	Trizma	MS/MSD
DW-240409-04	Finished water	09 Apr 10:05	09 Apr 16:20	Trizma	Corrected
DW-240409-05	Finished water	09 Apr 10:20	09 Apr 16:20	Trizma	Routine

Packet order is controlled; supporting bench records and field-level amendments remain part of the audit trail.

Calibration model, transitions, and limits

Quantifier and qualifier transitions; reporting limits apply to the original sample

$$R = A_{\text{native}} / A_{\text{IS}}$$

$$C_{\text{sample}} = ((R - b) / m) \times \text{DF}$$

$$\text{Recovery \%} = ((C_{\text{spiked}} - C_{\text{native}}) / C_{\text{added}}) \times 100$$

$$U95 = 2 \times \sqrt{(u_{\text{cal}}^2 + u_{\text{rep}}^2 + u_{\text{vol}}^2 + u_{\text{rec}}^2)}$$

TRANSITION AND REPORTING-LIMIT TABLE

Identity		Transitions		Retention	Validated range	Limits
Analyte	Internal standard	Quantifier	Qualifier	RT min	Range ng/L	LOD / LOQ ng/L
PFBA	13C4-PFBA	213>169	213>119	2.18	0.50-80	0.16 / 0.50
PFPeA	13C5-PFPeA	263>219	263>169	2.74	0.50-80	0.13 / 0.50
PFHxA	13C5-PFHxA	313>269	313>119	3.32	0.50-80	0.18 / 0.50
PFHpA	13C4-PFHpA	363>319	363>169	3.86	0.50-80	0.15 / 0.50
PFOA	13C8-PFOA	413>369	413>169	4.41	0.25-80	0.08 / 0.25
PFNA	13C9-PFNA	463>419	463>169	4.95	0.25-80	0.09 / 0.25
PFDA	13C6-PFDA	513>469	513>219	5.48	0.25-80	0.10 / 0.25
PFBS	18O2-PFBS	299>80	299>99	3.05	0.50-100	0.17 / 0.50
PFHxS	18O2-PFHxS	399>80	399>99	4.63	0.50-100	0.20 / 0.50
PFOS	13C8-PFOS	499>80	499>99	5.64	0.50-100	0.22 / 0.50
6:2 FTS	13C2-6:2FTS	427>407	427>81	5.19	1.00-100	0.31 / 1.00
HFPO-DA	13C3-HFPO-DA	329>285	329>169	3.71	0.50-80	0.14 / 0.50

QUALIFIER CONTRACT

Symbol	Meaning	Rendering rule
U	Not detected at the LOQ	Preserve the leading < sign and LOQ
J	Estimated concentration	Preserve the numeric value and J
-	No qualifier	Do not invent a textual qualifier

Calibration curves and residual disposition

Six-analyte weighted review | fit, residual, and exclusion disposition



METHOD ACCEPTANCE RULE

Use weighted $1/x^2$. Accept $R^2 \geq 0.995$ and each included back-calculated standard within +/-15%; the lowest included level may be within +/-20%. A standard may be excluded only under a signed, source-visible exception.

Panel-specific fit, exclusion, and residual disposition are recorded in Figure 03; the signed calibration exception log controls exclusions.

Table V4 - Accuracy and precision

Part 1 of 2 | fortified reagent water | recovery / RSD in percent

Analyte	Low rec / RSD %	Mid rec / RSD %	High rec / RSD %	n	State
PFBA	96.0 / 8.4	101.0 / 4.6	99.0 / 3.8	7	Pass
PFPeA	93.0 / 9.1	98.0 / 5.2	101.0 / 4.1	7	Pass
PFHxA	91.0 / 10.3	97.0 / 5.8	100.0 / 4.3	7	Pass
PFHpA	95.0 / 7.9	99.0 / 4.9	102.0 / 4.0	7	Pass
PFOA	98.0 / 6.8	103.0 / 3.7	101.0 / 3.2	7	Pass
PFNA	97.0 / 7.2	102.0 / 4.1	100.0 / 3.6	7	Pass

STUDY DESIGN AND ACCEPTANCE

Level	Nominal basis	Replicates	Recovery	RSD
Low	LOQ or 2 x LOQ	7 on 7 days	70-130%	<=20%
Mid	10 ng/L	7 on 7 days	70-130%	<=20%
High	80 ng/L	7 on 7 days	70-130%	<=20%

Continuation boundary: PFDA through HFPO-DA appear on page 5 under repeated column headers.

Table V4 - Accuracy and precision (continued)

Part 2 of 2 | repeated headers are structural, not data rows

Analyte	Low rec / RSD %	Mid rec / RSD %	High rec / RSD %	n	State
PFDA	92.0 / 11.5	96.0 / 6.3	99.0 / 4.9	7	Pass
PFBS	104.0 / 9.7	99.0 / 4.8	98.0 / 4.0	7	Pass
PFHxS	101.0 / 8.9	100.0 / 4.4	97.0 / 3.9	7	Pass
PFOS	89.0 / 12.6	95.0 / 6.7	96.0 / 5.1	7	Pass
6:2 FTS	86.0 / 14.8	92.0 / 7.4	94.0 / 6.3	7	Pass
HFPO-DA	99.0 / 7.5	104.0 / 4.2	103.0 / 3.5	7	Pass

EXCEPTION REGISTER LINKED TO TABLE V4

Record	Affected result	Observation	Disposition
EX-24-011	6:2 FTS low-level day 2	61% recovery	Excluded: fortification omitted
EX-24-014	PFOS low-level day 6	89% recovery; 12.6% RSD	Retained: within criteria
EX-24-017	HFPO-DA mid day 4	104% recovery	Retained: within criteria

EX-24-011 is not part of n=7. The valid replacement replicate is included in the 86.0% low-level mean.

Extraction, fortification, and dilution bench record

Working bench copy | handwritten checks transcribed at review

BENCH ORIGINAL

PREPARATION LEDGER

Prep ID	Material / sample	Initial	Aliquot	Final	Recorded target / DF
SPK-01	LCS mixed native spike	1.000 ng/uL	10.00 uL	1.000 L	10.00 ng/L
SPK-02	MS/MSD mixed native spike	1.000 ng/uL	10.00 uL	1.000 L	10.00 ng/L
DIL-03	DW-240408-03 extract	1.000 mL	0.500 mL	5.000 mL	DF 10.00
DIL-04	DW-240409-04 original extract	1.000 mL	1.000 mL	1.000 mL	DF 1.00
DIL-05	DW-240409-04 reinjection vial	0.800 mL	0.400 mL	0.800 mL	DF 2.00

EXTRACTION LEDGER

Sample	Bottle mL	Extract mL	SPE lot	Surrogate added	Analyst check
LRB-240410-01	250.0	1.000	SPE-24-118	50.0 uL	MI 10 Apr 08:04
LCS-240410-01	250.0	1.000	SPE-24-118	50.0 uL	MI 10 Apr 08:08
DW-240408-03	249.0	1.000	SPE-24-118	50.0 uL	MI 10 Apr 08:17
DW-240409-04	250.0	1.000	SPE-24-118	50.0 uL	MI 10 Apr 08:23

BENCH ANNOTATIONS

- [x] SPK-01 and SPK-02 verified against CRM certificate PFAS-MIX-2319, expiry 30 Sep 2024.
- [x] DIL-05 prepared only after reinjection authorization MX-24-041; keep DIL-04 vial in audit storage.
- [x] Balance check 10.0002 g against 10.0000 g; pipette P100 verification 99.6 uL at 20.1 C.

M. Iyer / analyst / 10 Apr 2024 08:42

L. Chen / witness / 10 Apr 2024 08:51

LC-MS/MS sequence and technical review

Batch 240410 | signed sequence sheet

Seq	Type	Vial	Sample / control	Start	IS rec	Carryover	State
09	Cal	A09	CAL-80	09:54	103%	Clear	Accepted
10	Blank	B01	LRB-240410-01	10:07	97%	Clear	Accepted
11	LCS	B02	LCS-240410-01	10:20	94%	Clear	Accepted
12	Sample	B03	FB-240408-01	10:33	96%	Clear	Accepted
13	Sample	B04	DW-240408-02	10:46	92%	Clear	Accepted
14	Sample	B05	DW-240408-03	10:59	89%	Clear	Accepted
15	MS	B06	DW-240408-03-MS	11:12	93%	Clear	Accepted
16	MSD	B07	DW-240408-03-MSD	11:25	90%	Clear	Accepted
17	CCV	B08	CCV-10	11:38	95%	Clear	Accepted
18	Sample	B09	DW-240409-04	11:51	38%	Clear	HOLD MX-24-041
19	Sample	C01	DW-240409-04-RI	12:18	91%	Clear	Accepted
20	Sample	C02	DW-240409-05	12:31	88%	Clear	Accepted
21	CCV	C03	CCV-10-CLOSE	12:44	93%	Clear	Accepted

TECHNICAL REVIEW ANNOTATIONS

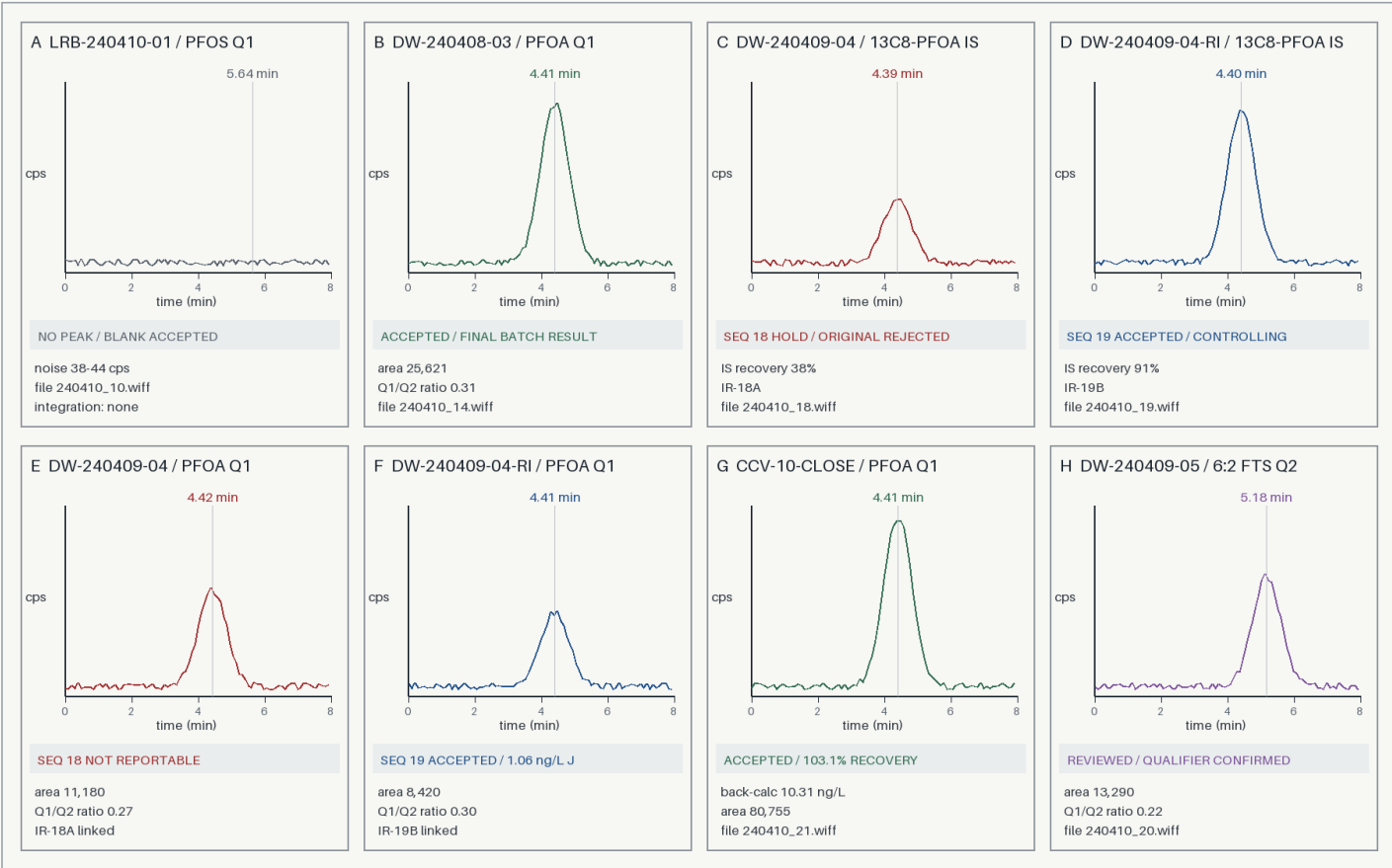
Linked item	Entry
MX-24-041	Opened 11:58 after sequence 18 IS recovery fell below 50%
Authorization	One reinjection after blank and needle-seat flush
Control rule	If reinjection IS is 50-150%, retain original only in audit trail
Review mark	Sequence 19 circled; sequence 18 struck once, still legible

M. Iyer / analyst / 19 Apr 2024 14:05

L. Chen / technical / 22 Apr 2024 09:40

Raw integration and chromatogram review

Eight-panel quantifier, qualifier, and internal-standard review



INTEGRATION CROSS-REFERENCE INDEX

Panel	Injection	Trace channel	Review role
A	Seq 10	PFOA Q1	Reagent blank
B	Seq 14	PFOA Q1	Batch sample
C	Seq 18	13C8-PFOA IS	Original internal standard
D	Seq 19	13C8-PFOA IS	Reinjection internal standard
E	Seq 18	PFOA Q1	Original quantifier
F	Seq 19	PFOA Q1	Reinjection quantifier
G	Seq 21	PFOA Q1	Closing CCV
H	Seq 20	6:2 FTS Q2	Qualifier review

The index cross-references injection and trace channel. Signed instrument annotations are the controlled source for retention time, area, ratio, recovery, file, and review state.

Blank, recovery, and continuing-calibration review

Measured values shown here; fortification targets are controlled by the page 6 bench record

Control	Analyte	Measured	Review criterion	Review state
LRB-240410-01	PFOA	<0.25 ng/L U	<0.25 ng/L	Meets
LRB-240410-01	PFOS	<0.50 ng/L U	<0.50 ng/L	Meets
LCS-240410-01	PFOA	9.62 ng/L	70-130% recovery	Calculate from SPK-01
LCS-240410-01	PFOS	9.14 ng/L	70-130% recovery	Calculate from SPK-01
DW-240408-03-MS	PFOA	13.02 ng/L	70-130% recovery	Subtract native 3.42
DW-240408-03-MSD	PFOA	13.22 ng/L	RPD <=30%	Compare with MS
DW-240408-03-MSD	6:2 FTS	8.40 ng/L	70-130% recovery	Native <1.00 U
CCV-10	PFOA	9.48 ng/L	85-115% of 10.00	Meets
CCV-10-CLOSE	PFOA	10.31 ng/L	85-115% of 10.00	Meets

REVIEWER CALCULATION INDEX

Computation	Required source join	Reviewer action
LCS PFOA recovery	p6 SPK-01 target + p9 measured	Calculate; compare to 70-130%
LCS PFOS recovery	p6 SPK-01 target + p9 measured	Calculate; compare to 70-130%
MS PFOA recovery	p6 SPK-02 + native + p9 spiked	Subtract native; calculate
MSD PFOA recovery	p6 SPK-02 + native + p9 spiked	Subtract native; calculate
MS/MSD PFOA RPD	two calculated spike recoveries	Calculate; compare to <=30%
MSD 6:2 FTS recovery	p6 SPK-02 + p9 measured; native U	Treat native as zero; calculate

The calculation index gives routing and review actions; the signed preparation record remains the only target source.

Instrument maintenance and reinjection authorization

Maintenance exception MX-24-041 | service bench record

ORIGINAL SIGNED RECORD

Time	Observation / action	Measured check	Initials
11:55	Sequence 18 flagged by IS rule	13C8-PFOA recovery 38%	MI
12:00	Pressure and source check	6,820 psi; spray stable	MI
12:04	Needle-seat inspection	Visible salt film	LC
12:08	Needle seat flushed 3 x 500 uL	50:50 MeOH:water	LC
12:12	System blank injected	PFOS carryover <0.08 ng/L	MI
12:16	IS check solution	13C8-PFOA recovery 96%	MI
12:17	One reinjection authorized	Use retained DIL-05 vial	LC

AUTHORIZATION AND BOUNDARY

Decision field	Signed entry
Permitted scope	DW-240409-04 only; one reinjection
Acceptance gate	IS recovery 50-150%; blank carryover below LOQ
If gate passes	Send both integrations to QA; do not select final concentration here
If gate fails	Invalidate sample result and request re-extraction
Audit preservation	Retain sequence 18, IR-18A, and original vial DIL-04

HANDWRITTEN SERVICE NOTE

Salt residue was confined to the needle seat. No source cleaning, recalibration, or re-extraction was authorized.

M. Iyer / operator / 19 Apr 2024 12:22

L. Chen / authorization / 19 Apr 2024 12:24

LIMS export and signed QA correction

Provisional export snapshot with a separately signed correction record

PRELIM-LIMS-0419 PROVISIONAL EXPORT

Export row	Sample	Analyte	Result	Qual.	Source	Status
E-104	DW-240409-04	PFOA	1.84 ng/L	J	Seq 18 / IR-18A	Provisional
E-105	DW-240409-04	PFOS	<0.50 ng/L	U	Seq 18 / IR-18A	Provisional
E-106	DW-240409-05	PFOA	4.87 ng/L	-	Seq 20 / IR-20A	Ready

SIGNED CORRECTION RECORD

QUALITY CORRECTION RECORD CR-24-044

CONTROLLED / SIGNED

Scope: DW-240409-04 only. Other batch results are unchanged.

Superseded source: PRELIM-LIMS-0419 entries derived from sequence 18 / IR-18A.

Controlling source: accepted reinjection sequence 19 with integration IR-19B.

Correct PFOA from 1.84 ng/L J to 1.06 ng/L J.

PFOS remains <0.50 ng/L U; no qualifier change is authorized.

Reason: sequence 18 internal-standard recovery was below the 50-150% criterion.

Effective after technical and QA signatures below; do not overwrite the audit trail.

Technical: L. Chen / 22 Apr 2024 09:40

QA: R. Patel / 22 Apr 2024 16:10

Linked records: SEQ-240410, IR-18A, IR-19B, MX-24-041

Final signed validation and batch determination

Release values after source-precedence review | 22 Apr 2024

Result ID	Sample	Analyte	Released result	Qual.	U95	Authority
R-01	FB-240408-01	PFOA	<0.25 ng/L	U	-	Routine
R-02	DW-240408-02	PFOA	2.18 ng/L	-	0.42 ng/L	Routine
R-03	DW-240408-02	PFOS	0.62 ng/L	J	0.18 ng/L	Routine
R-04	DW-240408-03	PFOA	3.42 ng/L	-	0.55 ng/L	Routine
R-05	DW-240408-03	PFOS	0.74 ng/L	J	0.19 ng/L	Routine
R-06	DW-240409-04	PFOA	1.06 ng/L	J	0.25 ng/L	CR-24-044
R-07	DW-240409-04	PFOS	<0.50 ng/L	U	-	CR-24-044
R-08	DW-240409-05	PFOA	4.87 ng/L	-	0.72 ng/L	Routine

DETERMINATION

Release check	Determination	Basis
Calibration	Accepted	Weighted fits and residual review
Blanks and CCVs	Accepted	Opening and closing controls within criteria
Recovery / precision	Accepted	Valid n=7; documented EX-24-011 replacement
DW-240409-04	Accepted as corrected	Sequence 19 / IR-19B / CR-24-044
Batch release	RELEASED	J and U qualifiers retained exactly

Release boundary: the signed determination does not erase provisional exports, rejected integrations, or maintenance records.

L. Chen / technical / 22 Apr 2024 09:40

R. Patel / QA release / 22 Apr 2024 16:10